Beulah Mercy Paul Manickam

STAFF SOFTWARE ENGINEER

Highly accomplished Staff Software Engineer with nearly 12 years of experience in architecting, developing, and deploying cutting-edge software solutions. Proficient in full-stack development, cloud computing, and microservices architecture, with a proven track record of leading successful software projects from conception to completion. Skilled in Agile methodologies, including biweekly sprint models, scrums, and retrospectives, ensuring continuous delivery and improvement. Known for strong problem-solving abilities, effective communication, and a focus on optimizing performance and cost efficiency. Seeking a challenging role to contribute innovative solutions and drive team success in dynamic environments.

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SKILLS

- Programming Languages: Java 11, Python, Node.js, HTML
- Frameworks: Spring Boot, Spring MVC, Spring Data, Spring REST, JPA, Hibernate, Mockito, React
- Technologies: RESTful Web Services, GraphQL API, JAX-RS, ActiveMQ, Docker
- Database: MySQL, Oracle 10g
- Cloud Services (AWS): Lambda, EC2, ECS, S3, CloudFront, Route 53, CloudWatch, RDS, DynamoDB, AppSync, API Gateway, AppConfig, Step Functions, SQS, SNS, SAM, VPC
- Cloud Services (Azure): APIM, Functions, Kubernetes Services (AKS), Application Insights, Cosmo DB
- IDEs: IntelliJ IDEA, Spring Tool Suite, Visual Studio Code
- Version Control: Git, BitBucket
- Build Tools: Maven, Gradle
- Agile Tools: JIRA
- CI/CD Tools: Jenkins, AWS CodeBuild, Azure DevOps

EXPERIENCE

Staff Software Engineer

Trimble Information Technologies India Pvt Ltd *Sep 2020 - Present*

Responsibilities:

- **Strategic Architecture Leadership:** Led enterprise-wide microservices transformation initiatives across multiple platforms, collaborating with architects to design scalable distributed systems that improved fault isolation and enabled independent service deployment for 50,000+ users across global construction and engineering teams.
- **Cross-Platform Development Management**: Managed end-to-end development lifecycle for critical business applications including Trimble Connect Platform and WorksManager, coordinating technical deliverables across multiple engineering teams while ensuring consistent architectural patterns and coding standards.
- **APICloud Services Development**: Designed and developed APICloud services within Trimble Cloud platform, creating enterprise API management and provisioning capabilities that serve construction applications, implementing API gateway solutions, rate limiting, and authentication frameworks supporting 2M+ daily API requests across the Trimble ecosystem.
- **Performance & Cost Optimization**: Drove systematic performance improvement initiatives across cloud infrastructure and application layers, implementing cost-effective solutions that achieved significant infrastructure savings while maintaining enterprise-grade performance and reliability standards.
- **Quality Assurance & Engineering Excellence**: Established comprehensive quality frameworks including automated testing strategies, code review processes, and production monitoring systems that reduced errors and downtime while maintaining high development velocity across distributed teams.
- **Team Leadership & Mentorship:** Provided technical leadership and mentorship to 10+ engineers across multiple projects, fostering professional growth through knowledge sharing sessions, code reviews, and hands-on guidance on enterprise development best practices and emerging technologies.

- **Agile Delivery Management:** Managed sprint planning, delivery timelines, and cross-team coordination to consistently meet business objectives and project milestones, implementing continuous improvement processes through regular retrospectives and stakeholder feedback integration.
- **Production Support & Incident Management:** Established proactive monitoring and incident response frameworks ensuring high system availability and rapid resolution of production issues, maintaining service level agreements and customer satisfaction across enterprise applications.

Senior Associate Projects

Cognizant Technology Solutions India Pvt. Ltd *Jan 2017 - Sep 2020*

Responsibilities:

- **Requirement Analysis**: Analyzed customer requirements and produced documentation, including understanding and technical solution documents, to meet enterprise application needs.
- **Solution Design and Architecture:** Assisted architects in designing backend solutions and API integration strategies, ensuring compliance with enterprise standards and design patterns.
- **Documentation and Communication:** Demonstrated and communicated design concepts through flowcharts, sequence diagrams, and pseudo-code, ensuring that project stakeholders fully understood the solutions
- **Modular Code Development:** Ensured modular code development for each component, making them easily testable and minimizing interdependencies.
- **Quality Assurance:** Led code reviews to ensure quality and adherence to best practices, enhancing the performance and security of solutions.
- **Production Issue Resolution:** Addressed and resolved production issues, performing debugging and collecting necessary information to facilitate efficient issue resolution.

Software Developer

D4t4 Solutions PLC *Aug 2015 - Dec 2016*

Associate Programmer

Aparajitha Corporate Services Private Limited *Oct 2014 - Jul 2015*

Web Developer

Native Sparrow Software Solutions *Oct 2013 - Sep 2014*

PROJECTS

Trimble WorksManager

Staff Sotware Engineer *May 2025 - Present*

Client: Civil Construction Field Systems (CCFS)

Technologies: Java 11, Spring Boot, Spring Data JPA, Spring Security, MongoDB, MySQL, Docker, Azure DevOps, Azure Boards, Azure Kubernetes Service (AKS), Grafana, Loki, React, Maven, Gradle

Platform Scale:

- 4,000+ active users
- 1,500+ active projects
- 70,000+ daily API requests

Technical Contributions:

- **Microservices Architecture & Development:** Designed and developed multiple Spring Boot microservices including Profile Service, Subscription Service, and core WorksManager services deployed on Azure Kubernetes Service (AKS), achieving 99.9% uptime with comprehensive authentication, authorization, and cross-service integration using REST APIs and event-driven messaging patterns.
- **Database Optimization & Performance**: Enhanced MongoDB and MySQL schemas across all services with compound indexing strategies and query optimization, reducing API response times from 800ms to 120ms and database load by 60% while supporting multi-tenant data isolation.
- **Authentication Framework & API Development**: Built centralized OAuth 2.0/JWT authentication system across all microservices, developed 50+ REST endpoints following OpenAPI 3.0 specifications with standardized validation and error handling achieving sub-200ms response times.

- **Integration Solutions & Caching**: Implemented Trimble Connect integration layer for project synchronization and user role management, deployed Redis distributed caching improving page load times from 2.3s to 0.8s, established event-driven architecture for seamless service communication.
- **DevOps & Kubernetes Deployment**: Established comprehensive test automation framework achieving 90%+ code coverage, designed Azure DevOps CI/CD pipelines integrated with Azure Boards for automated work item tracking and AKS deployments, implemented containerized Spring Boot microservices with Docker and Kubernetes orchestration for auto-scaling and rolling updates.
- **Technical Leadership & Code Quality**: Mentored 5+ engineers on microservices best practices, established coding standards and reduced technical debt by 35% through systematic refactoring initiatives, conducted code reviews and implemented quality gates aligned with development standards.
- **Monitoring & Observability**: Integrated Grafana/Loki logging with Prometheus metrics for centralized monitoring across AKS clusters, implemented production feedback loops for continuous improvement and performance optimization ensuring consistent system reliability.

Non-Functional Achievements:

- Performance: 40% memory reduction, 50% throughput improvement through JVM optimization
- Security: OWASP compliance with TLS 1.3 encryption and comprehensive input validation
- Scalability: Horizontal scaling support handling 10x traffic spikes during peak usage
- Reliability: 99.9% system availability through circuit breaker patterns and health monitoring
- Agile Excellence: Maintained 90%+ sprint goal achievement rate across multiple sprints with consistent team velocity

Trimble Connect Platform

Staff Software Engineer *May 2025 - Present*

Client: Trimble Construction Partners (Project Sight, Sketchup, etc.,)

Technologies: Java 17, Spring Boot 3.x, Spring Data JPA, Spring Security, Jackson 2.15, MySQL 8.0, DynamoDB, Redis, AWS (ECS, RDS, Lambda, S3, CloudWatch, API Gateway, CloudFormation), Docker, GitHub Actions, GitHub Workflows, Gradle 7.6, Maven, React, OpenAPI 3.0, Postman, ProfileX Integration, JIRA

Technical Contributions

- **Microservices Development:** Designed and developed 5 core Spring Boot microservices (Auth, User, Project, File, Permission) deployed on AWS ECS with GitHub Actions CI/CD, achieving 99.95% uptime and under 150ms response times across distributed services.
- **Database Optimization:** Enhanced MySQL and DynamoDB schemas across 15+ services with compound indexing and single-table design patterns, reducing API response times from 1.2s to 180ms and database load by 65% through query optimization and Redis caching.
- **Authentication System**: Built enterprise OAuth 2.0/JWT authentication supporting user and application identities, developed 80+ REST endpoints with OpenAPI 3.0 specifications, implemented delegation token handling achieving under 150ms authentication response times.
- **Performance Engineering:** Implemented Redis distributed caching improving session retrieval from 800ms to 50ms, optimized JWT processing reducing CPU usage by 30%, designed horizontal scaling supporting 15x traffic spikes with 99.9% cache hit rates.
- **Cloud Infrastructure**: Established AWS architecture with ECS containerization, RDS Aurora for MySQL, DynamoDB for NoSQL, S3/CloudFront for file storage, designed GitHub workflows for automated deployments with CloudFormation infrastructure as code.
- **Cross-Service Integration**: Implemented ProfileX integration for centralized identity management, established event-driven architecture using AWS SNS/SQS, designed Jackson polymorphic serialization with backward compatibility for external automation scripts.
- **Technical Leadership**: Mentored 8+ engineers across multiple teams on microservices patterns, conducted code reviews reducing technical debt by 40%, established coding standards and quality gates aligned with enterprise security compliance.
- **Production Excellence**: Integrated AWS CloudWatch monitoring for distributed services, implemented JIRA-based incident management with automated alerting, maintained 99.95% platform availability through circuit breaker patterns and comprehensive error handling.

Trimble APICloud Management v2.1

Lead Software Engineer *Sep 2020 - Apr 2024*

Client: Trimble Internals and External Partners

Technologies: Java 11, Spring Boot, Spring REST, Spring Data JPA, MySQL, Docker, Jenkins, AWS (SAM, Lambda, Layers, DynamoDB, Aurora, CodeBuild, SNS, DynamoDB Streams), Gradle, React JS, Azure APIM, Python

Contributions:

- Successfully migrated the RDS schema to a **DynamoDB single-table design**, optimizing for scalability and faster data retrieval, which improved both performance and disaster recovery (DR) readiness by distributing data across regions.
- Designed and implemented a **new authorization layer** using **OPA (Open Policy Agent)** to enforce resource-level access control, with robust filtering based on user roles and permissions, ensuring data security and integrity.
- Applied **Domain-Driven Design (DDD)** principles to define aggregates and value objects within bounded contexts, enhancing the modularity and maintainability of the system while ensuring business logic consistency.
- Developed **AWS SAM templates** for provisioning infrastructure, enabling seamless scaling, high availability, and rapid deployment during disaster recovery scenarios.
- Architected the solution with **auto-scaling** capabilities using AWS Lambda and DynamoDB, ensuring the system scaled effortlessly to handle increased loads without compromising performance.
- Implemented **Command Query Responsibility Segregation (CQRS)** by using **GraphQL** for efficient query handling (GET operations) and AWS API Gateway with Lambda for mutation endpoints (POST/PUT operations), ensuring optimized performance and faster response times.
- Managed authentication using **OAuth tokens**, providing secure and seamless access while maintaining resilience across services.
- Designed **disaster recovery** plans by leveraging multi-region DynamoDB replication and **SNS** for reliable notifications, ensuring system resilience and minimizing downtime.
- Contributed to the **design documentation**, creating both high-level and low-level architecture diagrams, including database schema designs, to ensure clarity in system resilience, integrity, and scalability.
- Played an integral role in **performance tuning** by optimizing database queries, employing caching strategies, and monitoring performance metrics through **AWS CloudWatch**, ensuring application resilience under high traffic.
- Actively participated in **requirement analysis**, breaking down epics into manageable user stories, identifying dependencies and blockers, and ensuring the project aligned with disaster recovery, resilience, and performance goals.
- Provided mentorship to senior and junior engineers, delegating tasks, conducting **code reviews**, and leading technical sessions on best practices for coding standards, resilience strategies, and performance optimization.

EDUCATION

Master of Computer Application

Velammal College of Engineering and Technology *2009 - 2012*

Percentage - 83.1%

Bachelor of Science (Computer Science)

Fatima College *2006 - 2009*

Percentage - 87.5%

Higher Secondary School Certificate

T.E.L.C Girls Higher Secondary School *2006*

Percentage - 80%

Secondary School Leaving Certificate

T.E.L.C Girl Higher Secondary School *2004*

Percentage - 94.4%

AWARDS

Star Award

Cognizant *2018-07-24*

For excellent team players and on time delivery with high quality.

DECLARATION

I hereby declare that the above-mentioned information is correct up to my knowledge and I bear the responsibility for the correctness of the above-mentioned particulars. If there is any vacancy in your company, I would like to apply. If I'm appointed, I can assure you to do my best and satisfy you with my work and the benefit of the organization

Place: Chennai

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